

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

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**COURSE NAME: Cryptography and Network Security**

**COURSE CODE: CSA5112**

### **CONCEPT MAPPING**

**1. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:**

**i.Show differences and similarities**

**ii.Map how each affects the Data structure and Security level**

**iii.Analyze which is more vulnerable to attacks and why**

**Answer**

Main Topic

Classical Encryption Schemes

Branch 1: Substitution Techniques

- Replaces one character with another.
- Position of letters remains the same.
- Changes the identity of characters.
- Examples:
  - o Caesar Cipher
  - o Monoalphabetic Cipher
  - o Playfair Cipher
  - o Vigenère Cipher

Branch 2: Transposition Techniques

- Rearranges the positions of characters.
- Characters remain unchanged.
- Changes the order of letters.
- Examples:
  - o Rail Fence Cipher
  - o Columnar Transposition Cipher

**3(a)(i) Differences and Similarities**

Similarities

- Both are classical encryption techniques.
- Both convert plaintext into ciphertext.
- Both require a key or encryption rule.
- Both aim to provide confidentiality.

Differences

Substitution Transposition

Changes letters Changes positions

Letter order remains same Letter values remain same

Character identity changes Character identity unchanged

Data Structure

Substitution

- Original order of letters is maintained.
- Characters are replaced with different letters.

- Example: HELLO → KHOOR

#### Transposition

- Letters remain the same.
- Order of letters changes.
- Example: HELLO → HLEOL

#### Security Level

##### Substitution

- Moderate security.
- Vulnerable to frequency analysis.

##### Transposition

- Better than simple substitution in hiding letter patterns.
- Still vulnerable if the transposition method is known.

#### 3(a)(ii) Which is More Vulnerable?

##### Substitution Technique

More vulnerable because:

- Letter frequencies remain unchanged.
- Attackers can use frequency analysis.
- Common words can be guessed easily.

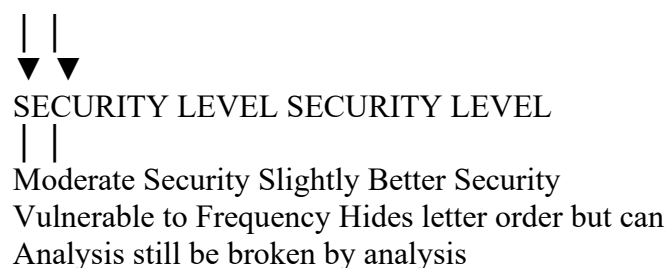
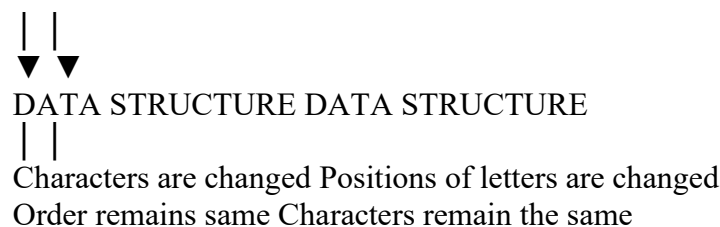
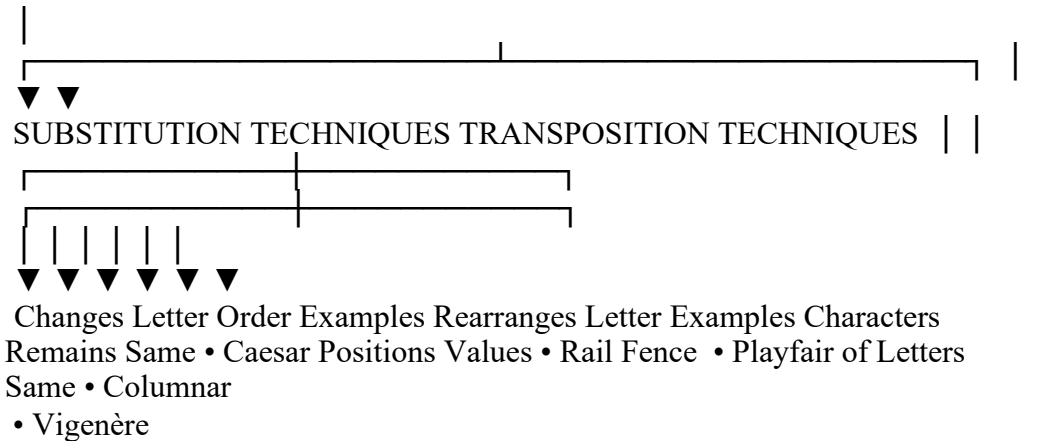
##### Transposition Technique

- Slightly more secure than simple substitution.
- Harder to identify word patterns.
- Still vulnerable to brute-force attacks and pattern analysis.

#### Conclusion

Substitution techniques are generally more vulnerable because the frequency of letters is preserved, making cryptanalysis easier. Transposition techniques provide slightly better security by hiding the order of letters, but they are also not secure enough for modern communication.

#### CLASSICAL ENCRYPTION SCHEMES



## SIMILARITIES

- Classical Encryption Methods
- Convert Plaintext to Ciphertext
- Use an Encryption Rule or Key
- Provide Confidentiality

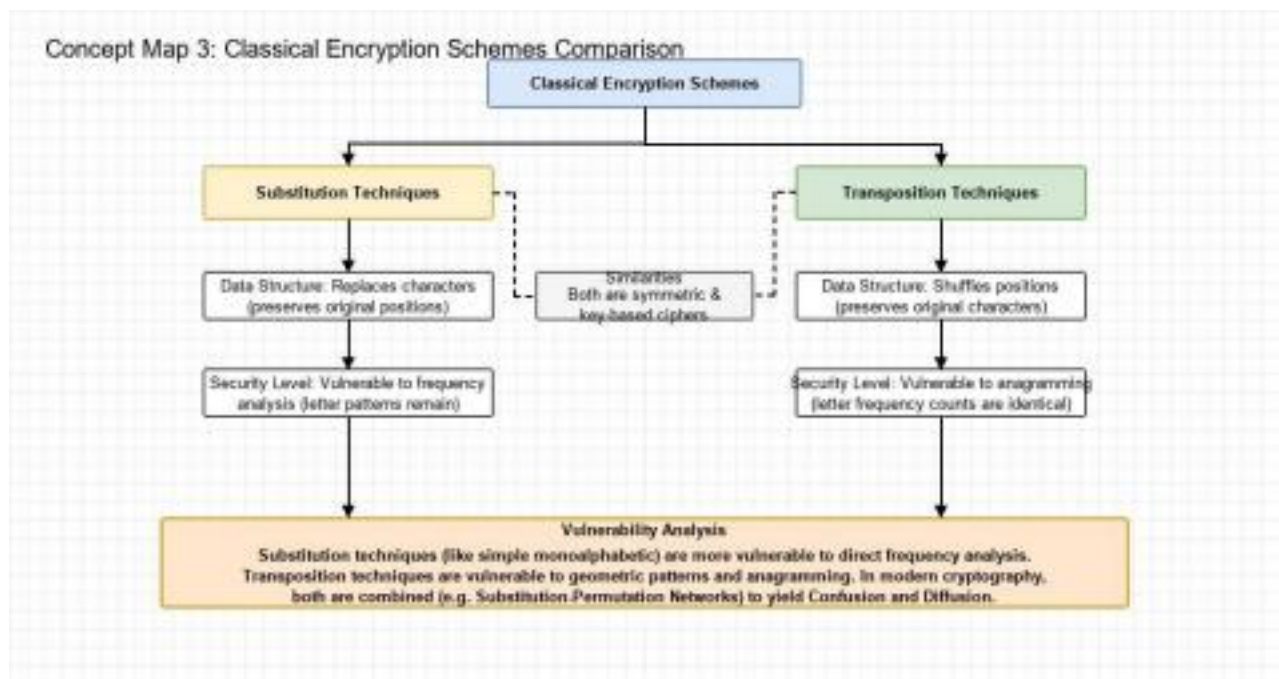
## VULNERABILITY ANALYSIS

### Substitution Techniques

- More vulnerable to attacks
- Frequency analysis is possible
- Common letter patterns remain

### Transposition Techniques

- Less vulnerable than substitution
- Letter order is hidden
- Still breakable if the method is known



**4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:**

- Show relationships between these concepts**
- Analyze how each contributes to secure communication**
- Include one real-world application for each**

## Answer

### Main Topic

#### Steganography

Steganography is the technique of hiding secret information inside another file such as an image, audio, video, or document so that the existence of the message is concealed. It is widely used for secure communication and copyright protection.

#### Data Hiding

- Conceals secret information inside digital media.

- Hides text, images, audio, or video.
- Makes communication invisible to unauthorized users.

Real-World Application:

- Secret military communication using images.

Watermarking

- Embeds ownership or copyright information into digital content.
- Protects intellectual property.
- Can be visible or invisible.

Real-World Application:

- Copyright protection for digital photos, videos, and documents.

Survivability

- Ensures hidden information remains even after compression, editing, or transmission.
- Measures the robustness of hidden data.
- Prevents loss of embedded information.

Real-World Application:

- Secure transmission of medical images and confidential government documents.

4(a) Relationships Between These Concepts

- Steganography is the main technique for hiding information.
- Data Hiding is the process of concealing secret information.
- Watermarking protects ownership and authenticity of digital content.
- Survivability ensures the hidden information remains intact after modifications.
- Together, these concepts improve secure and reliable communication.

4(b) Contribution to Secure Communication

Data Hiding

- Conceals confidential information from unauthorized users.
- Makes secret communication difficult to detect.

Watermarking

- Protects digital content from unauthorized copying.
- Verifies ownership and authenticity.

Survivability

- Maintains hidden information after compression, cropping, or editing.
- Ensures reliable and secure data transmission.

4(c) Real-World Applications

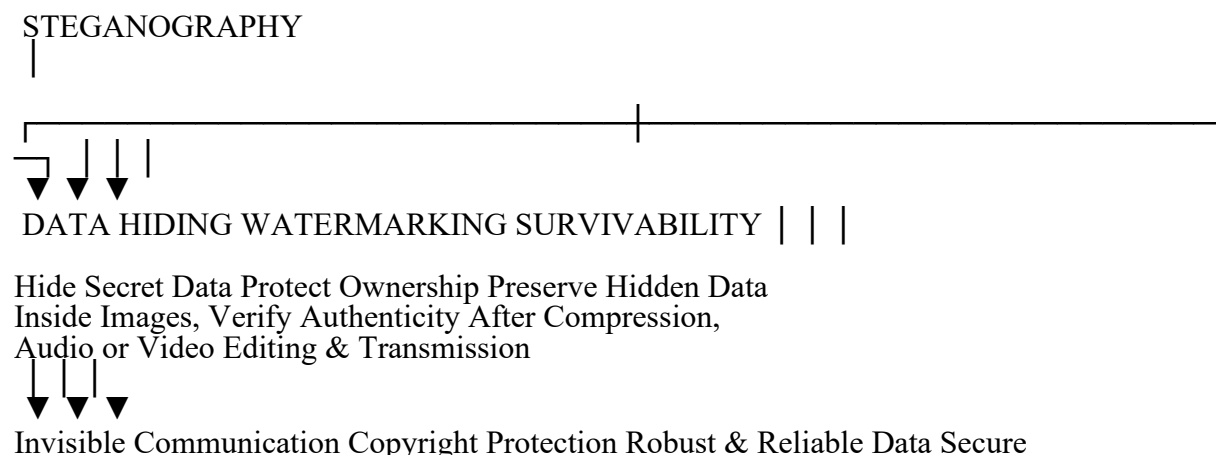
Concept Real-World Application

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Data Hiding Secret military communication using digital images

Watermarking Copyright protection for digital photos, videos, and documents

Survivability Secure transmission of medical images and government documents



## Communication Digital Authentication Recovery



Application Application Application  
Secret Military Copyright for Medical Images &  
Communication Photos, Videos Government Documents



## SECURE COMMUNICATION

- Confidentiality
- Authentication
- Integrity
- Reliable Data Transmission

